

PDW 4.1 Beta

Paging Decoder for Windows

Installation, decoding, filtering, notification, secure transfer and web publishing

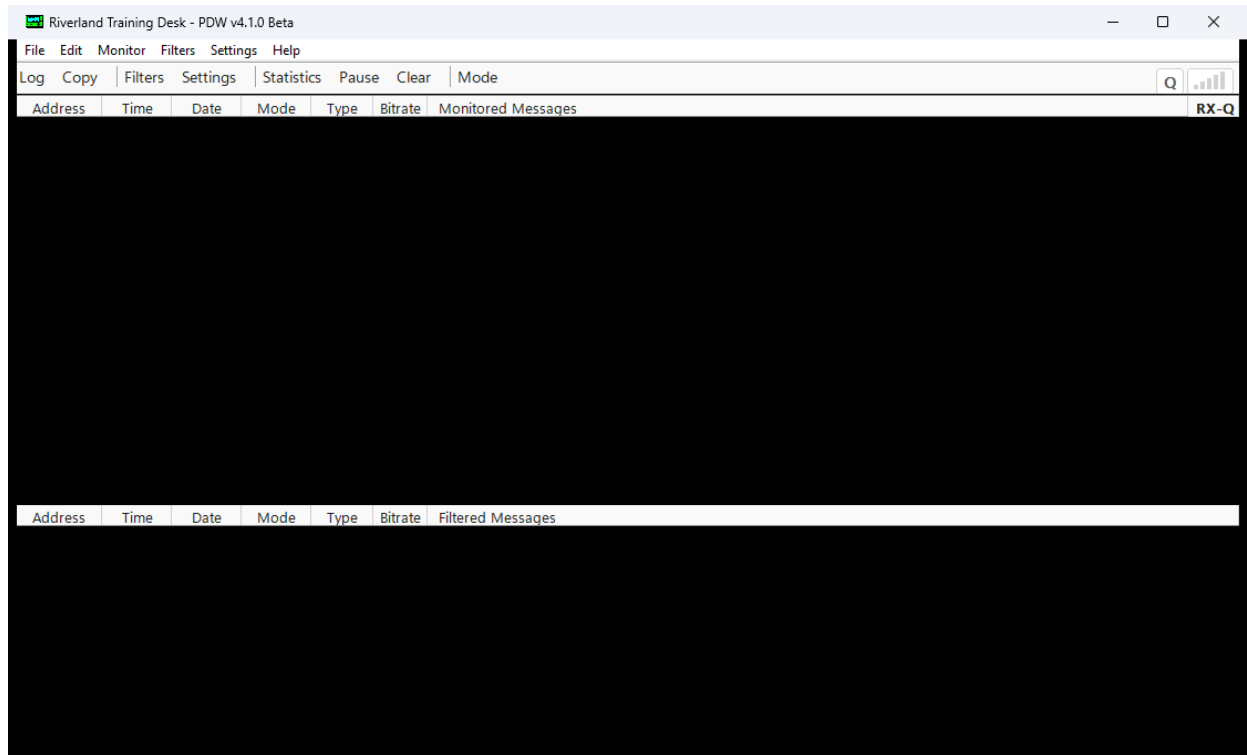


Figure 1. PDW 4.1 Beta task-oriented main window.

Edition 1.0 | August 2026 | Covers PDW v4.1.0 Beta

Portable 32-bit Windows application for Windows 10 and Windows 11

All messages, accounts, hostnames and credentials shown in this manual are fictional training data.

How to use this guide

Training data: Every pager address, message, aircraft registration, network identifier, hostname, account name, token and password in this guide is synthetic. Do not copy example secrets into a live installation.

This manual replaces the 2010 PDW v3.1 manual for everyday operation of the PDW v4.1.0 beta. It retains the original protocol, message-format, filter, logging and display guidance while updating the workflow for the current six-menu interface and the new receiver, notification, transfer and publishing features.

Feature status used in this manual

Status	Meaning
Available in 4.1 Beta	Present in the supplied PDW v4.1.0 Beta executable and interface.
Preserved legacy	Existing behaviour intentionally retained for compatibility.
Future / not included	Discussed only to prevent users mistaking roadmap work for a current feature.

Ten-minute first run

1. Extract the whole PDW folder to a writable location such as C:\PDW-Training. Do not run the executable from inside a ZIP file.
2. Start PDW v4.1.0 Beta.exe and open Settings > Settings.
3. Choose Follow Windows, Light or Dark, then open Decoder and input > Input setup.
4. Select local soundcard, serial input, RTL-TCP, direct RTL-SDR USB, or a recording replay. Use only a signal you are permitted to receive.
5. Choose Monitor > Decoding mode and select POCSAG/FLEX, ACARS, MOBITEK or ERMES.
6. Check the signal bar and receive-quality indication. Correct input level, gain, bandwidth and tuning before changing decoder thresholds.
7. Open Filters > Manage filters and create one synthetic training filter.
8. Enable logging only after choosing a folder with suitable privacy and retention controls.
9. Configure email, Apprise, transfer or publishing only after creating the required third-party account or service.
10. Send a configuration-only test, then confirm an unfiltered synthetic message does not trigger filtered-only outputs.

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1. Read this first

What PDW does

PDW is a native Windows signal decoder. Combined with a suitable receiver or interface, it can monitor supported POCSAG, FLEX, ACARS, MOBITEK and ERMES signals; show decoded records; filter selected records; log, alert and publish operator-approved outputs; and provide statistics and signal diagnostics.

Mode	Rates / role	Current boundary
POCSAG	512, 1200 and 2400 baud	Legacy parser retained; audio and serial sources supported.
FLEX	1600, 3200 and 6400 baud	Legacy parser retained; enhanced four-level audio adds B/D phase decisions beside A/C.
ERMES	6250 baud	Legacy parser retained.
ACARS	Aircraft data messages	Optional parity checking; database files can add labels and aircraft information.
MOBITEK	8000 baud in normal use	Base/mobile sync, network frame sync and message-type controls retained.

What is new in 4.1 Beta

Area	Available now	Compatibility / safety boundary
Windows interface	Six task-oriented menus, text toolbar, Settings hub, System/Light/Dark themes, DPI-aware dialogs.	Decoder algorithms and existing configuration files are retained.
Startup	Optional WindowTitle and Start PDW with Windows.	Automatic startup waits five seconds; one instance per application folder.
Audio	WinMM first, event-driven WASAPI fallback and bounded device-loss retry.	Legacy unsigned 8-bit path remains available.
Radio	Reconnecting RTL-TCP and direct RTL-SDR USB through a bundled 32-bit receiver pack.	Optional receiver DLLs load only after selection.
Diagnostics	WAV/SigMF record and replay, waveform, quality history, metrics and calibration.	Previous live source is restored after replay.
Secure transfer	FTP, explicit FTPS, implicit FTPS and password-authenticated SFTP.	FTPS validates certificates; SFTP requires a SHA-256 host key.
Push	Filtered-only Apprise delivery with test, background queue and privacy-safe default body.	Apprise API is operated separately; secrets use Credential Manager.
Publishing	Static JSON/JSONL/RSS/Atom/HTML and generic HTTPS webhooks.	Disabled by default; jurisdiction acknowledgement and published-copy privacy controls are required.

Features still being added or held behind evidence gates

Not a current feature	Current position
Second independent decoder / soft-decision FEC	Not enabled. Representative licensed or redacted recordings and before/after evidence are required first.
SoapySDR	Not included. The narrower RTL-TCP and librtlsdr paths are the supported modern

Not a current feature	Current position
	radio routes.
MQTT and publishing batches	Possible future extensions. Current static files and HTTPS JSON webhook are complete without them.
64-bit PDW	Not included. The executable remains 32-bit to retain legacy serial and slicer compatibility.

Privacy and law: PDW cannot determine whether reception, storage, forwarding or publication is lawful in your location. Check permissions, radio rules, privacy obligations and publication rules before use. Treat decoded content as potentially private.

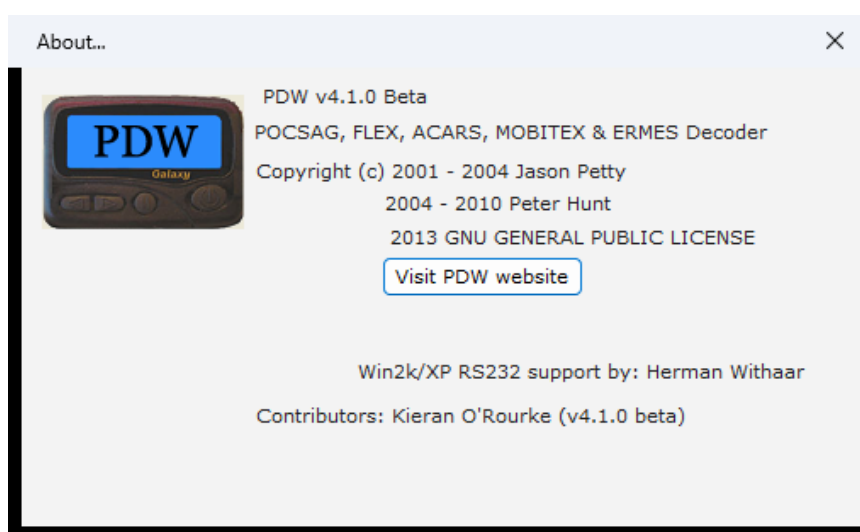


Figure 2. About dialog for the supplied PDW v4.1.0 beta.

2. Install and first start

Requirements

Item	Requirement
Operating system	Windows 10 or Windows 11.
Application	PDW v4.1.0 Beta.exe and the folders supplied with it.
Architecture	32-bit application; it runs on normal 64-bit Windows but receiver DLLs imported into PDW must be compatible 32-bit librtlsdr builds.
Input	A Windows audio device, supported serial/RS232 interface, RTL-TCP server, compatible RTL-SDR USB receiver, or a supported WAV/SigMF replay file.
Network	Required only for external notification, email, transfer, webhook or network receiver features.

Portable installation

11. Create a normal writable folder, for example C:\PDW-Training.
12. Extract the complete release into that folder. Keep Receivers, docs, PDW.INI, filters.ini and the executable together.
13. If Windows shows a download warning, verify the release source and checksum before running it. Do not bypass a warning for an unverified copy.
14. Start PDW v4.1.0 Beta.exe. The default source remains the legacy local input and publishing features remain off.
15. Close PDW normally after the first configuration change so the INI files are written cleanly.

Do not mix releases: Keep each portable PDW installation in its own folder. Copying only the new executable over an incomplete old folder can omit receiver packages, documentation or dependency files.

Upgrade from an older PDW folder

16. Close the old PDW instance.
17. Copy PDW.INI, filters.ini, Wavfiles and any required database files to a private backup location.
18. Keep Logfiles and custom recordings according to your retention policy; they may contain decoded private content.
19. Extract the beta to a new folder, then copy only the settings and data you intentionally want to migrate.
20. Confirm input, filters, logging and output paths before enabling automation.

Name separate instances

A separate folder can monitor a separate receiver or network. Add a WindowTitle value under [PDW] so the main window and tray tooltip identify the instance.

```
[PDW]
WindowTitle=Riverland Training Desk
```

The resulting title is Riverland Training Desk - PDW v4.1.0 Beta. A single folder still permits only one running instance.

Start with Windows

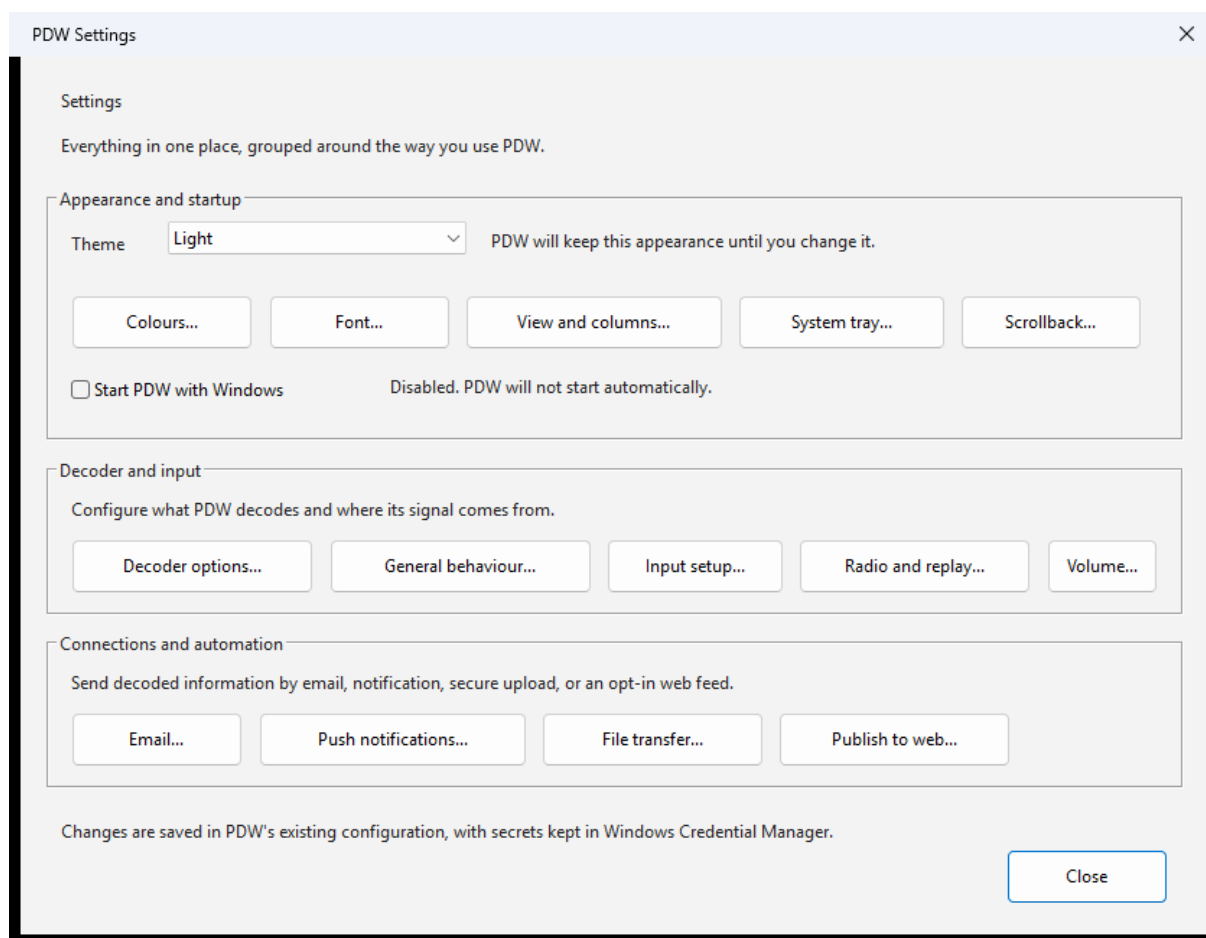


Figure 3. Central Settings hub with appearance, startup, input and automation groups.

Open Settings > Settings and select Start PDW with Windows. Automatic launches wait five seconds after sign-in to allow audio devices and services to settle. Clearing the same box disables the startup entry. Manual launches remain immediate.

Important files and folders

Path	Purpose / privacy
PDW.INI	Main settings. External passwords and tokens for modern features are not written here.
filters.ini	Filter definitions, labels, hit counters and related options.
Wavfiles	Optional WAV alerts, including address- and text-specific files.
Logfiles	Operator logs. Protect, retain and delete according to policy.
Receivers	Bundled and imported optional receiver packages plus the Zadig driver tool.
Published	Default static publishing output. May contain transformed decoded events.
PublishQueue / DeadLetter	Pending and repeatedly failed webhook events. Treat as sensitive.
FileTransfer.log / Apprise.log	Sanitized operational results. Apprise.log excludes endpoints, credentials and decoded content.

3. Interface tour

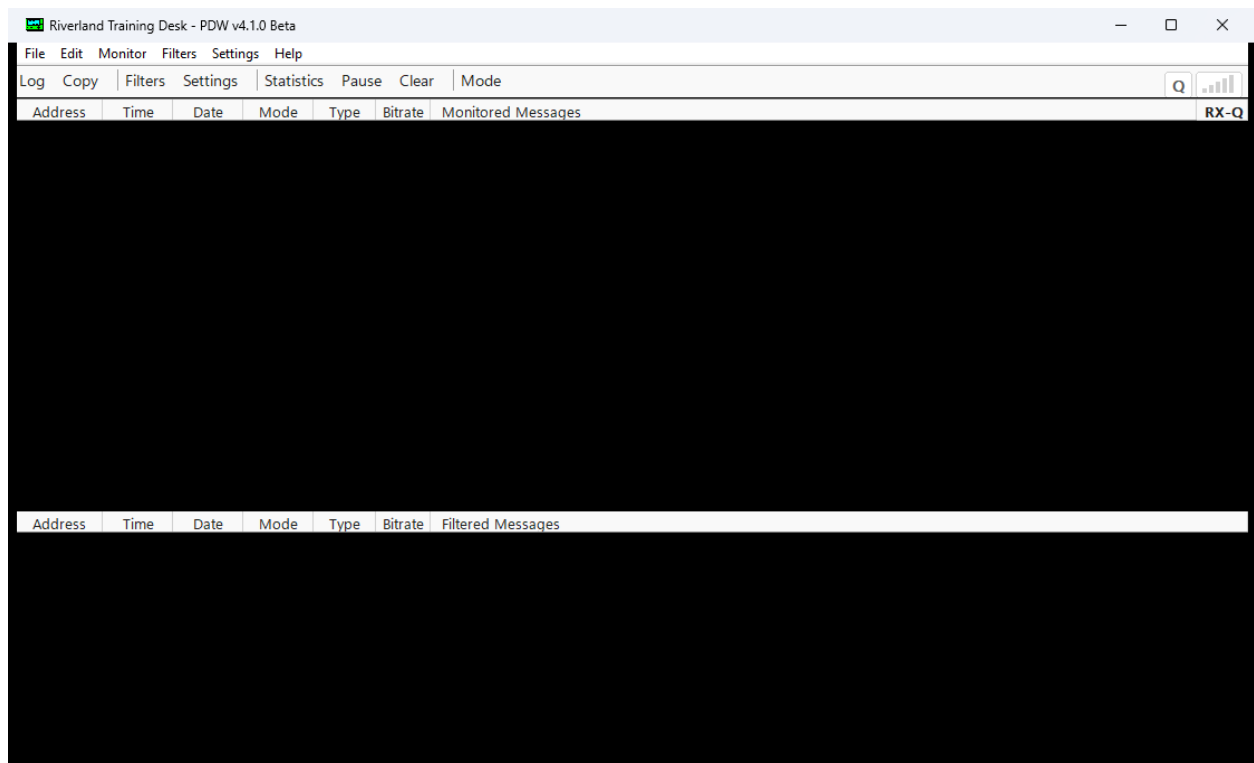


Figure 4. Main window in the Light appearance.

Window areas

Area	Use
Menu bar	All commands, organised as File, Edit, Monitor, Filters, Settings and Help.
Text toolbar	One-click Log, Copy, Filters, Settings, Statistics, Pause, Clear and Mode actions.
Monitor pane	Upper pane containing displayed decoded messages.
Filtered pane	Lower pane containing messages that meet active non-reject filters and are not Monitor Only.
Divider	Drag to change the relative pane sizes; F11 switches pane sizing.
Signal indicator	Activity and receive-quality indication. It is not a calibrated RF power meter.

The six menus

Menu	Main tasks
File	Open or close a log, save copied data, print copied data, exit.
Edit	Copy selection, monitor window or filtered window; clear monitor panes.
Monitor	Input setup, volume, decoder mode, statistics, pause or resume.
Filters	Manage and reload filters, filter options, counters and command/file toggles.
Settings	Appearance, decoder/input, email, push, transfer, publishing and character set.
Help	User guide, diagnostics and About.

Appearance and sizing

Follow Windows changes with the system appearance. Light and Dark keep the chosen appearance. The current beta uses Segoe UI for modern dialogs, current Windows common controls and per-monitor DPI awareness. Legacy message colours and monospaced message fonts remain configurable.

Accessibility: Use Settings > Appearance > Font for a readable monospaced message font, View and columns to remove unneeded fields, and Scrollback to balance history against performance. Keyboard shortcuts remain available when a mouse is inconvenient.

Toolbar actions

Button	Action
Log	Open or close the selected log file.
Copy	Copy selected decoded text.
Filters	Open the filter manager.
Settings	Open the central Settings hub.
Statistics	View decoder statistics.
Pause	Pause or resume display/decoding operation as provided by PDW.
Clear	Choose which monitor pane to clear.
Mode	Change the current decoding mode.

4. Choose a signal source

Source decision guide

Source	Use it when	Important notes
Local audio	A receiver supplies discriminator, line, tape, speaker or earphone audio.	WinMM is tried first; WASAPI is an automatic fallback. Audio input is the simplest modern route for many receivers.
Serial slicer / RS232	You have compatible legacy hardware and drivers.	Two- and four-level slicer and RS232 behaviour is retained. Do not install obsolete VXD drivers on current Windows.
RTL-TCP	An RTL-TCP-compatible receiver runs locally or on a trusted network host.	PDW receives IQ, configures the tuner, demodulates NFM and reconnects. Decoded text is not sent to the server.
Direct RTL-SDR USB	A compatible RTL2832U/V3/V4/V4L receiver is attached to this PC.	Uses the bundled 32-bit standard pack. The exact USB interface may require WinUSB through Zadig.
Replay	You are diagnosing a licensed, redacted or synthetic recording.	Replays through the normal protocol routines and restores the previous live source afterward.

Legacy local audio and serial

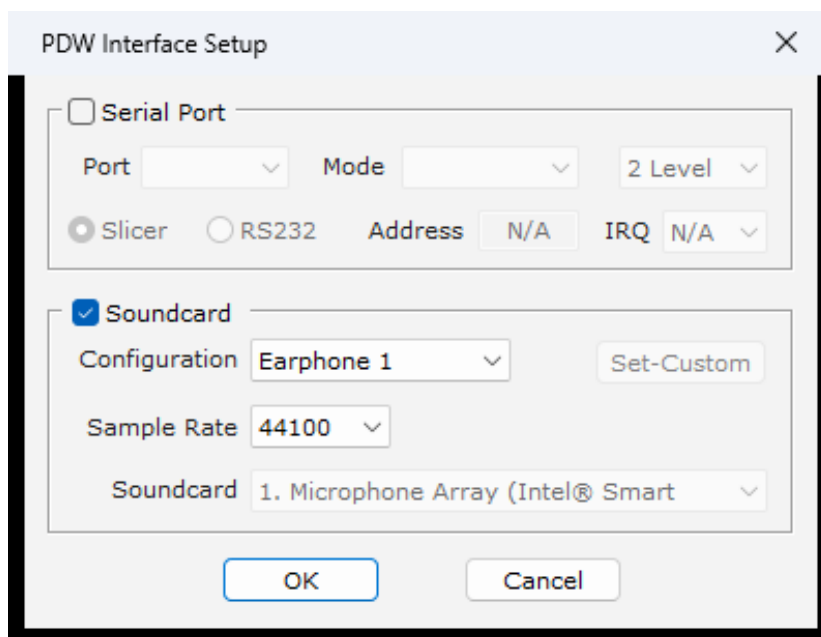


Figure 5. Legacy Input Setup remains available inside the modern interface.

21. Connect the receiver output or compatible serial interface before opening PDW.
22. Open Monitor > Input setup or Settings > Decoder and input > Input setup.
23. For audio, select Soundcard, choose the device and start with the closest source preset, normally Discriminator or Earphone.
24. For serial, select the exact COM port, Slicer or RS232, and the required level/mode. Current COM numbers above 9 are supported.
25. Select the matching decoder mode, tune to permitted activity and watch the signal indicator.
26. Adjust receiver/input level so the signal is visible without persistent clipping. If the legacy format cannot open, WASAPI fallback is automatic.

Receiver modification: A discriminator tap can improve decoding but requires receiver-specific technical work. Use a qualified technician or the correct service documentation; an incorrect modification can damage equipment or defeat electrical safety.

Direct RTL-SDR USB

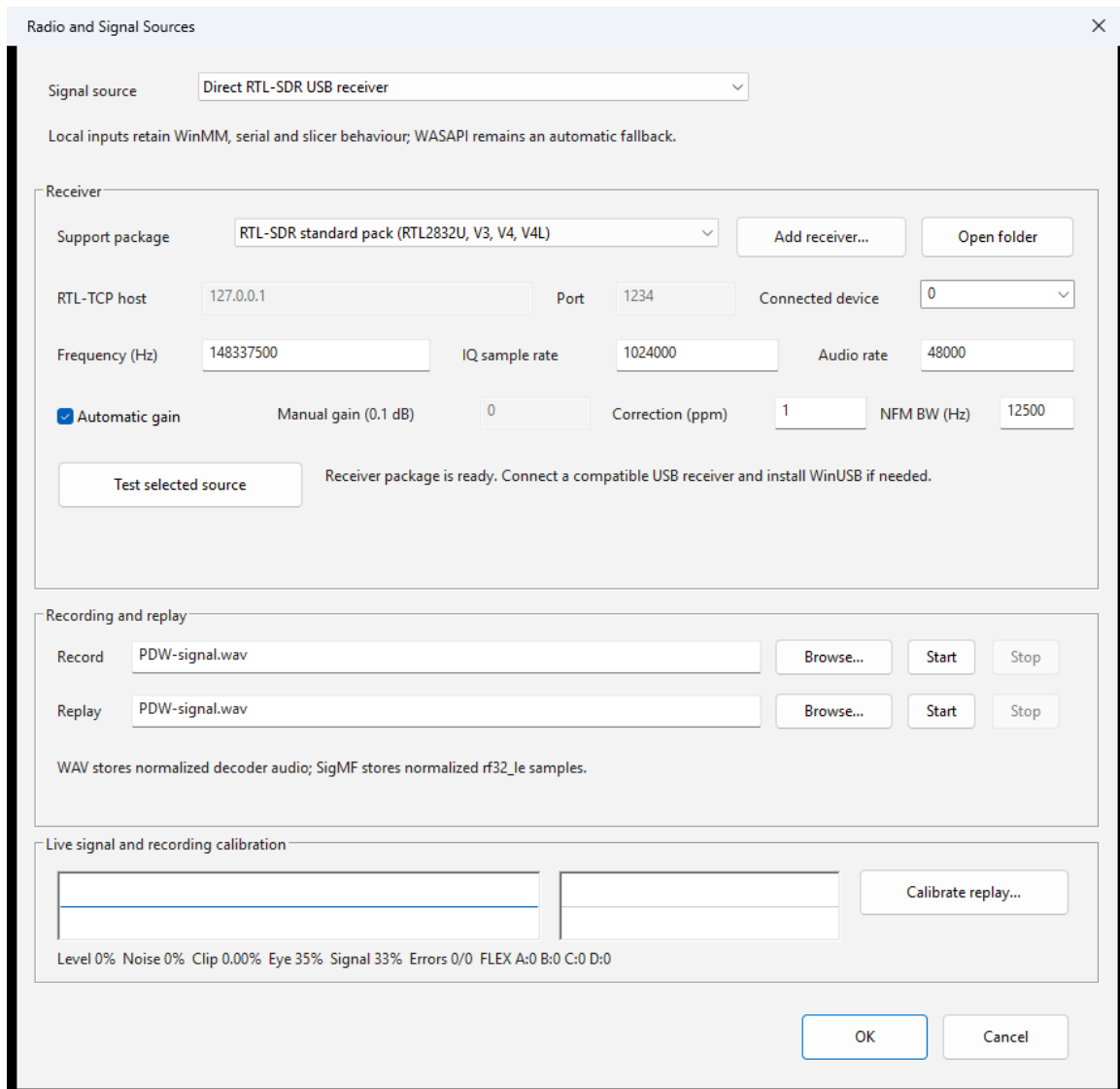


Figure 6. Direct RTL-SDR USB source with the bundled receiver pack.

The release includes a 32-bit standard receiver package for common RTL2832U receivers and RTL-SDR Blog V3, V4 and V4L devices. Open folder shows the package. Add receiver imports a trusted 32-bit rtlSdr.dll or librtlSdr.dll and neighbouring dependencies into Receivers\Custom after architecture and API checks.

Install WinUSB only when required

PDW includes Zadig 2.9 under Receivers\Driver Tools. The authoritative download and usage page is [Zadig - USB driver installation made easy](#).

27. Disconnect unrelated non-essential USB devices.
28. Run Zadig as administrator and choose Options > List All Devices only if the RTL-SDR interface is not visible.
29. Select the receiver's Bulk-In Interface 0, often shown as RTL2838UHIDIR or a similar RTL name. Confirm the USB ID against the device documentation.
30. Select WinUSB, then install or replace the driver.

31. Return to PDW, choose Direct RTL-SDR USB receiver, select the connected device and use Test selected source.

Driver warning: Selecting a keyboard, mouse, storage device, hub or unrelated receiver in Zadig can replace that device's driver and stop it working. PDW never launches Zadig or changes drivers automatically.

For current vendor-specific V3/V4 setup guidance, see the [RTL-SDR Blog Windows quick start guide](#).

RTL-TCP network receiver

Radio and Signal Sources

Signal source: RTL-TCP compatible network receiver

Local inputs retain WinMM, serial and slicer behaviour; WASAPI remains an automatic fallback.

Receiver

Support package: RTL-SDR standard pack (RTL2832U, V3, V4, V4L) [Add receiver... Open folder]

RTL-TCP host: decoder-gateway.example.net Port: 1234 Connected device: 0

Frequency (Hz): 148337500 IQ sample rate: 1024000 Audio rate: 48000

☒ Automatic gain Manual gain (0.1 dB): 0 Correction (ppm): 1 NFM BW (Hz): 12500

Test selected source Receiver package is ready. Connect a compatible USB receiver and install WinUSB if needed.

Recording and replay

Record: C:\PDW-Training\Recordings\2026-08-10-test-signal.wav [Browse... Start Stop]

Replay: C:\PDW-Training\Recordings\synthetic-flex-training.wav [Browse... Start Stop]

WAV stores normalized decoder audio; SigMF stores normalized rf32_le samples.

Live signal and recording calibration

[Waveform display] [Calibrate replay...]

Level 0% Noise 0% Clip 0.00% Eye 35% Signal 33% Errors 0/0 FLEX A:0 B:0 C:0 D:0

OK Cancel

Figure 7. RTL-TCP example using realistic fictional training values.

Field	Training example	Meaning
RTL-TCP host	decoder-gateway.example.net	Hostname or IP of the trusted RTL-TCP listener.
Port	1234	Common RTL-TCP port; use the listener's actual port.
Frequency (Hz)	148337500	Whole-hertz tuner frequency. Use only an authorised target.
IQ sample rate	1024000	Samples per second received from RTL-TCP.
Audio rate	48000	Internal demodulated audio rate.
Automatic gain	On	Start here; use manual gain only when needed.
Correction (ppm)	1	Frequency correction for receiver oscillator error.

Field	Training example	Meaning
NFM BW (Hz)	12500	Adjustable from 5000 to 25000; start near the signal's occupied bandwidth.

Record, replay and calibrate

32.WAV recording stores normalized decoder audio as 16-bit mono.

33.SigMF recording stores normalized rf32_le samples plus sample-rate metadata.

34.Replay accepts PCM8, PCM16 or float32 WAV and real float32 SigMF as documented by the beta.

35.Recording observes normalized samples without changing the decoder input. The in-memory diagnostic safety limit is 25 million samples.

36.Calibrate replay evaluates all 1,000 legacy threshold, centering and resync combinations, then asks before applying its signal-based suggestion to Custom configuration.

Test-data rule: Use only synthetic, redacted or explicitly licensed recordings in bug reports, training packs or public repositories. Live pager traffic may contain private information.

5. Decode and improve a signal

Select the decoder mode

Use Monitor > Decoding mode or the Mode toolbar button. PDW monitors one family at a time: POCSAG/FLEX, ACARS, MOBITEK or ERMES. Selecting the wrong family can show activity without valid messages.

The screenshot shows the 'PDW Options - Currently monitoring : Paging (Pocsag/Flex)' dialog box. It is divided into several sections for different decoder families:

- POCSAG:**
 - ☒ Enable Pocsag Decoding
 - ☒ 512 ☒ 1200 ☒ 2400
 - ☐ Decode function numbers always as default (1=Numeric 2=Tone 3=Tone 4=Alphanumeric)
 - ☐ If no accurate guess is possible, display message both as numeric and alphanumeric
- FLEX:**
 - ☒ Enable Flex Decoding
 - ☒ 1600 ☒ 3200 ☒ 6400
 - ☒ Show Short Instructions ☒ Convert Short Instructions to Text ☐ Use FlexTIME (not detected)
- MOBITEK:**
 - Bit Sync: Bitrate: Frame Sync: Min. Characters:
 - ☒ Bit Scrambler ☒ Ramnet ☒ Check Frame Sync
 - ☒ Show Network Messages ☒ Show TEXT ☒ Show DATA ☒ Show HPDATA
 - ☐ Verbose (control messages) ☐ Show Sweep Frames (BaseInfo) ☐ Show HPID
- ACARS:**
 - ☐ Full Parity Checking ☒ No Parity Checking
- Optional information in title bar:**
 - ☐ FLEX / ERMES Cycles & Frames ☐ Rejected / Blocked Messages

At the bottom are buttons for 'OK', 'General', and 'Cancel'.

Figure 8. Decoder options while monitoring POCSAG/FLEX.

POCSAG

- 37.Enable POCSAG and the rates expected on the monitored system: 512, 1200 and/or 2400.
- 38.Decode function numbers as default only when the network uses the protocol function bits consistently.
- 39.Show both numeric and alphanumeric can help when the type cannot be inferred reliably, but it increases duplicate interpretations on screen.

FLEX

- 40.Enable the required 1600, 3200 and/or 6400 rates.
- 41.Short Instructions can be shown or converted when the network uses dynamic group calls.
- 42.FlexTIME appears only when detected. Review system-time implications before allowing any time correction.

MOBITEX

43. Choose Base or Mobile bit sync for downlink or uplink monitoring.
44. Start with Check Frame Sync when discovering the network identifier, then enter the confirmed Frame Sync for cleaner decoding.
45. Use the minimum character count and TEXT/DATA/HPDATA controls to reduce unreadable transparent traffic.

ACARS and ERMES

46. ACARS parity checking can suppress corrupted records but may hide traffic when input quality is poor. Compare results during setup.
47. ERMES uses its retained 6250-baud parser and is best matched to a suitable four-level input.

Improve reception in the right order

48. Confirm the correct decoder mode and a known-active, permitted signal.
49. Confirm the source is actually selected and Test selected source succeeds for RTL inputs.
50. Correct frequency and PPM, then set a reasonable NFM bandwidth.
51. Adjust antenna position and receiver gain. Avoid persistent clipping.
52. For local audio, try the supplied Discriminator, Earphone, Speaker Out and Tape/Rec Out presets before Custom.
53. Record a short synthetic or authorised sample and compare diagnostics.
54. Use Calibrate replay only after the source, level and tuning are correct, and review the suggestion before applying it.

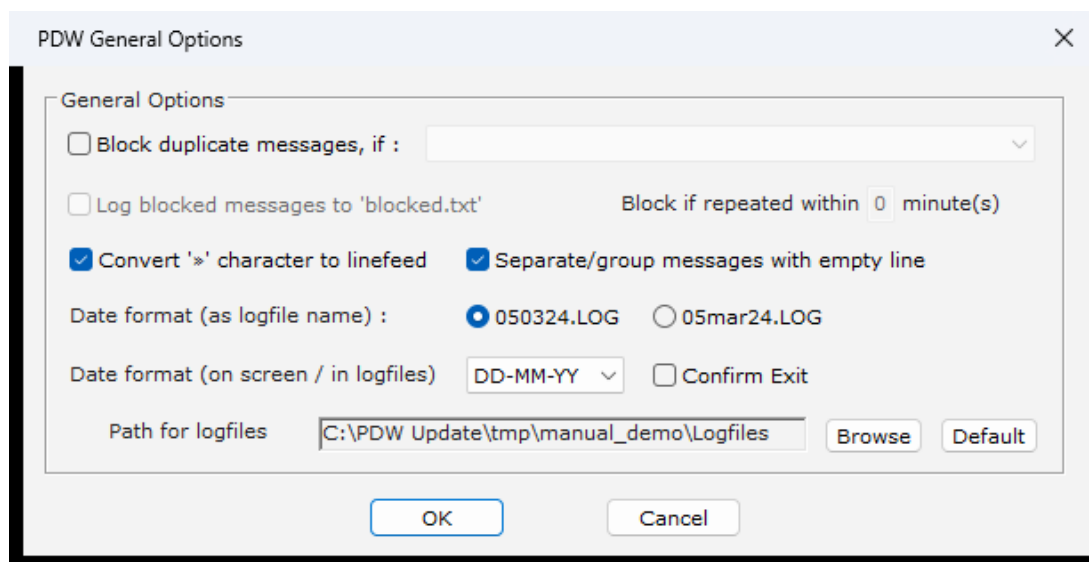


Figure 9. General behaviour settings for duplicate blocking, grouping, dates and log paths.

Duplicate blocking

Duplicate blocking can compare the previous address and message, the previous message only, or address/function/message within a timer. Enable blocked.txt during tuning if you need to review what is being suppressed. A blocked duplicate is also excluded from filtered-only push routing.

Signal indicator and new diagnostics

The signal indicator reacts to input activity and valid decoding; it is not calibrated RF signal strength. The Radio and Signal Sources dialog adds level, noise, clipping, eye opening, signal score, corrected versus uncorrectable error totals, FLEX A-D phase totals and a short receive-quality history. Use the combination, not one number, to judge changes.

Enhanced decoding boundary

The 4.1 beta runs adaptive DC and envelope tracking beside the legacy slicer. Four-level audio FLEX preserves the legacy sign decisions for phases A/C while adding inner/outer decisions for phases B/D; low-confidence input falls back to the legacy 0/3 path. The hard-decision protocol parser and serial paths remain authoritative. A second full parser and soft FEC are not present.

6. Understand decoded messages

Synthetic examples: The records in this chapter are formatted like PDW output but are invented. Addresses, registrations, flights, times and content do not identify a real person, aircraft or service.

Paging format: POCSAG, FLEX and ERMES

Address	Time	Date	Mode	Type	Rate	Message
1234567	14:13:33	10-08-26	POCSAG-1	ALPHA	1200	TRAINING DRILL - Unit 12 report to base
7654321	14:15:08	10-08-26	POCSAG-2	NUMERIC	512	0400000000
202345678	14:17:25	10-08-26	FLEX-A	ALPHA	1600	TEST ONLY - Airport operations standby
3456789	14:20:41	10-08-26	ERMES-2	ALPHA	6250	TRAINING WEATHER - exercise complete

Field	Meaning
Address	Pager or receiver address, normally seven or nine digits in the supported formats. Filters can match it exactly or with ? wildcards.
Time / Date	The receive timestamp using the computer clock unless a supported, detected network time option is deliberately used.
Mode	Protocol plus POCSAG function number, FLEX phase letter or ERMES designation.
Type	Alphanumeric, Numeric, Tone-only, Binary, Secure or Transparent as supported by the protocol.
Rate	The detected bit rate.
Message	Decoded content. Faulty characters, types and matched text may use configured colours.

ACARS format

Air.Reg	Time	Date	Msg.No	DBI	Mode	Message
VH-TST	16:42:07	10-08-26	F18A	4	R	H1 TRAINING DATA - DO NOT ACTION

ACARS item	Meaning
Aircraft registration	Registration reported in the ACARS record. Database enrichment may add country and aircraft type.
Message number	Four-character sequence indicator used by the network.
DBI	Downlink Block Identifier used to identify retransmissions.
Mode	Ground System Interface Configuration value.
Message label	Two-character label; label.df can add or update descriptions.
Flight / airline / station	Derived fields when the message and local database files provide enough information.
ACK / NAK	Technical acknowledgement or not-acknowledged indication.

MOBITEX format

MAN	Time	Date	Sender	Type	Rate	Message
7654321	15:14:13	10-08-26	7001200	TEXT	8000	TRAINING NETWORK TEST
1234567	15:16:41	10-08-26	7001201	HPDATA	8000	B+&(AB+&6 TEST

MOBITEX item	Meaning
MAN	Seven-digit Mobitex Access Number for the receiving radio modem.
Sender	Seven-digit sending MAN.
Type	TEXT, DATA, HPDATA or STATUS.
Rate	Normally 8000 in this PDW workflow.
Message	Raw 8-bit payload. Efficient application data can look random without being encrypted; the end-user application defines its meaning.

Choose visible columns

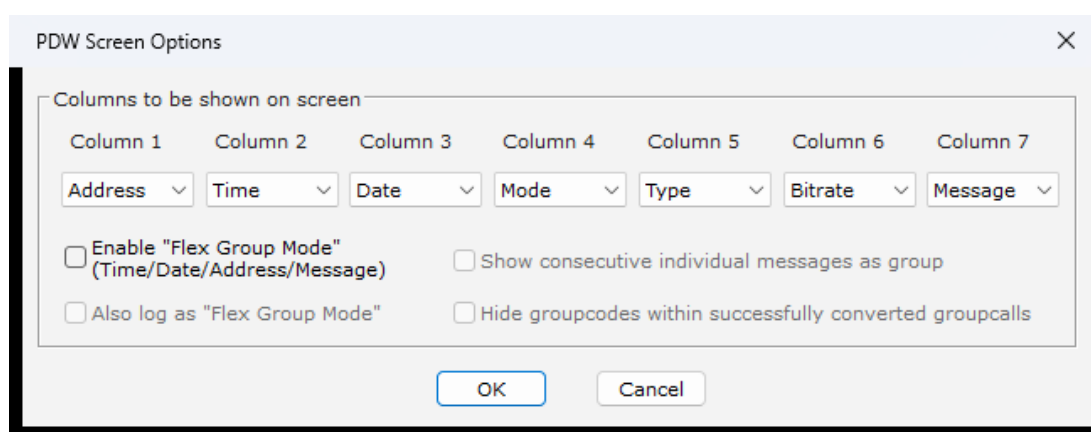


Figure 10. View and columns for the paging monitor.

Settings > Appearance > View and columns controls which fields appear and their order. FLEX Group Mode can combine group messaging where appropriate. Keep enough identifiers to interpret and audit records, but hide unused fields for a clearer display.

Character sets

Settings > Character set chooses Default ASCII or a character set loaded from language.df, such as German, Hebrew, Skyper or Swedish where supplied. Editing language.df adds entries after restart. Back up the file before editing and keep a known-good encoding.

7. Build filters and alerts

What a filter does

A filter compares an address and/or message text. A normal match can place a record in the filtered pane, add a label, play a WAV, write separate files, increment a hit counter and, when separately enabled, route email or Apprise. Reject filters hide matching records. Monitor Only matches can label or sound without entering the lower pane.



Figure 11. Filter manager with three fictional training filters.

Create a filter

55. Open Filters > Manage filters and choose Add.
56. Choose FLEX Riccode, POCSAG Riccode, Text, ERMES, ACARS or MOBITEK as applicable.
57. Enter the address or text rule. Use ? as a single-position address wildcard.
58. For a POCSAG address, choose All or a specific function number.
59. Add up to ten required text fragments separated by &, for example TRAINING&UNIT 12. Matching is not case-sensitive.
60. Add a clear label and select Show Filter Label if operators need it in output.
61. Choose Reject, Monitor Only, email, command file, sound and separate-file actions deliberately.
62. Choose OK, then use a synthetic input to confirm both a matching and a non-matching record.

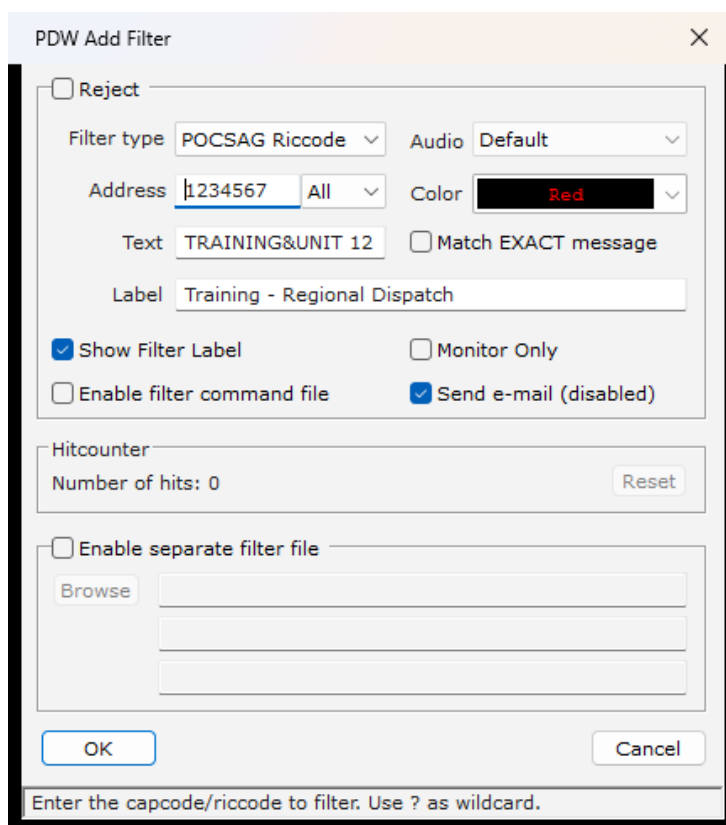


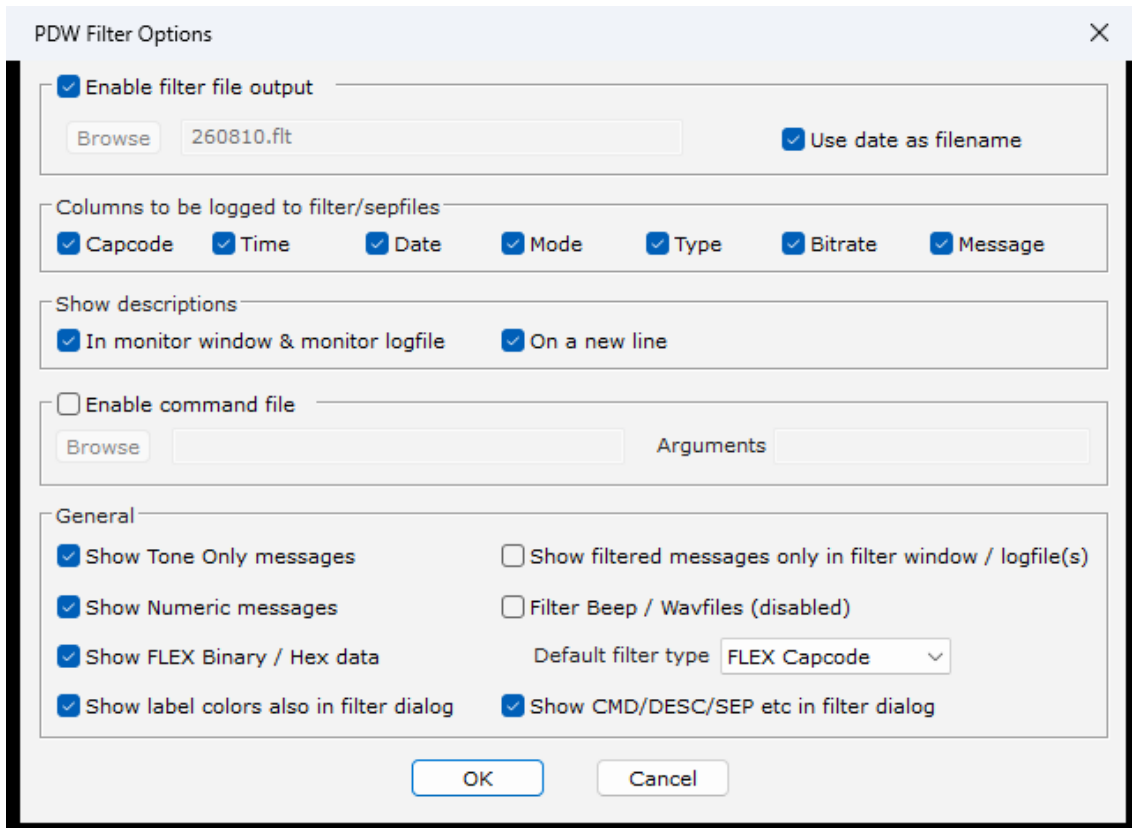
Figure 12. POCSAG filter using a realistic fictional address and text rule.

Field	Behaviour
Reject	Prevents matching messages from being displayed or routed as normal filtered output.
Filter type	Selects address/message interpretation for the active protocol.
Address	Exact address or ? wildcard pattern, such as 12?????.
Text	Case-insensitive phrase, exact-message rule or up to ten required fragments separated by &.
Label	Operator-facing description, optionally displayed and colour-coded.
Monitor Only	Allows a match action without placing the message in the lower filtered pane; it is not sent through filtered-only Apprise.
Send email	Active only when email is configured; use Selected filters only when each filter should decide.
Command file	Runs the configured external command for this filter. Treat decoded text as untrusted input.
Separate files	Writes up to three filter-specific files when filter-file output is enabled.
Hit counter	Tracks count and last hit; reset individual or selected/all counters as required.

Worked training rules

Goal	Type / address	Text	Expected result
One POCSAG address plus two words	POCSAG 1234567 / All	TRAINING&UNIT 12	Both fragments must appear in any case.
Address range	POCSAG 12?????	blank	Any seven-digit address beginning 12.
Exact test phrase	Text	TRAINING COMPLETE	With Match EXACT, longer messages do not match.
Suppress known test source	Appropriate address	TEST ONLY	Reject hides matching records.

Filter options



The PDW Filter Options dialog box is titled "PDW Filter Options" and includes a close button (X) in the top right corner. It contains several sections for configuring filter behavior:

- Enable filter file output:** A checked checkbox. Below it is a "Browse" button and a text field containing "260810.flt". To the right is a checked checkbox labeled "Use date as filename".
- Columns to be logged to filter/sepfiles:** A section with seven checked checkboxes: Capcode, Time, Date, Mode, Type, Bitrate, and Message.
- Show descriptions:** A section with two checked checkboxes: "In monitor window & monitor logfile" and "On a new line".
- Enable command file:** An unchecked checkbox. Below it is a "Browse" button, an empty text field, and a label "Arguments" followed by another empty text field.
- General:** A section containing:
 - Four checked checkboxes: "Show Tone Only messages", "Show Numeric messages", "Show FLEX Binary / Hex data", and "Show label colors also in filter dialog".
 - Two unchecked checkboxes: "Show filtered messages only in filter window / logfile(s)" and "Filter Beep / Wavfiles (disabled)".
 - A label "Default filter type" followed by a dropdown menu currently showing "FLEX Capcode".
 - A checked checkbox: "Show CMD/DESC/SEP etc in filter dialog".

At the bottom of the dialog are "OK" and "Cancel" buttons.

Figure 13. Filter file, display, sound and command options.

Filter Options controls a shared filtered logfile, date-based names, logged columns, description placement, message types, filter-window behaviour, sounds and the default filter type. Show filtered messages only in filter window/log files removes them from the upper pane but does not change the source record itself.

WAV alerts

The Default sound uses the chosen built-in WAV. Address-specific names can override it: 1234567.wav, or 1234567-fire.wav when address and text match. More specific address files take priority over wildcard names, which take priority over generic filtered and Monitor_only WAV files. Test volume outside a live incident workflow.

Command-file placeholders

Token	Value
%1	Address
%2	Time
%3	Date
%4	Mode
%5	Type
%6	Bitrate

Token	Value
%7	Message
%8	Label

Command safety: Decoded message text can contain unexpected characters. Quote arguments, send them to a fixed trusted program, avoid shell interpretation where possible, and never build privileged commands from pager content.

8. Log, copy, display and review

Main logfile

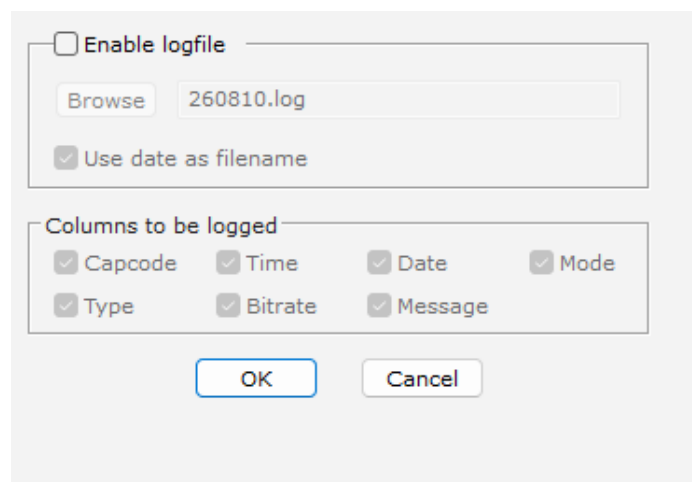


Figure 14. Logfile dialog with date-based naming and selected columns.

63. Open File > Log file or select Log on the toolbar.

64. Choose a protected output folder under Settings > General behaviour.

65. Enable the logfile, browse to a fixed file or select Use date as filename.

66. Select the minimum columns required for operations and audit.

67. Confirm the date format and write permissions, then inspect the first record.

Retention: A logfile can contain addresses, timestamps and message content. Limit access, back it up only when required, and define a deletion schedule. Sanitized transfer/notification logs are not a substitute for protecting the decoded log itself.

Copy, save and print

Command	Result
Copy Selection	Copies highlighted decoded text.
Copy monitor window	Copies all text in the upper pane.
Copy filtered window	Copies all text in the lower pane.
Save copied data	Writes the current clipboard copy to a text file.
Print copied data	Sends the current clipboard copy to the selected printer.

Display and operator comfort

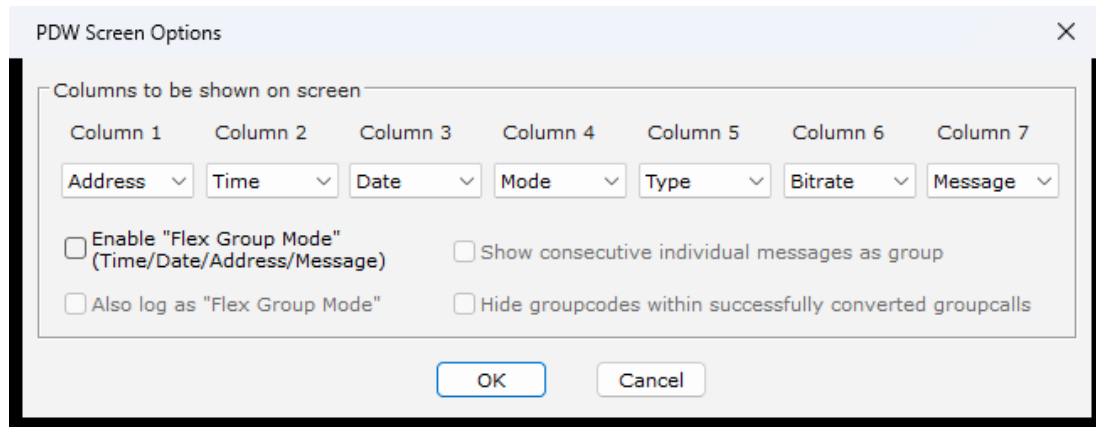


Figure 15. Columns and FLEX group display controls.

- 68.Colours opens a protocol-specific colour dialog for fields, filter matches, labels and bit errors.
- 69.Font accepts a non-proportional message font so columns remain aligned.
- 70.Scrollback controls monitor and filtered line history, mouse-wheel speed and pane sizing.
- 71.System tray can keep PDW running in the background and restore on configured message events.
- 72.Statistics shows recent message totals; use it with signal diagnostics rather than as a standalone quality measure.
- 73.Clear lets the operator clear either pane without deleting existing files already written to disk.

Generated output inventory

Output	Created by	Review point
Date or fixed .log	Main logfile	May contain all displayed messages.
Filter file(s)	Filter options / per-filter files	May contain selected private messages and labels.
blocked.txt	Duplicate blocking review	Use temporarily and inspect before deletion.
WAV / SigMF	Diagnostic recording	Signal data can still carry private content.
messages.* / index.html	Static publishing	Contains the published-copy fields selected by the operator.
FileTransfer.log	Continuous transfer	Operational results; password is excluded.
Apprise.log	Push delivery	Sanitized event/result entries; decoded text and destinations are excluded.

9. Email and push notifications

Third-party accounts: PDW does not create email, Apprise, ntfy or Pushover accounts. Set up and test the external service first, then enter only service-specific credentials in PDW. Terms, pricing and limits can change.

Email with an SMTP relay

Figure 16. Filtered email example using SMTP2GO settings and a fictional account.

SMTP2GO setup example

Create an account at [SMTP2GO signup](#), then create a dedicated SMTP user under Sending > SMTP Users.

74. Verify the sender domain or sender address in the provider dashboard.
75. In PDW open Settings > Connections and automation > Email.
76. Select Enable Email Notification and choose All messages, Filtered messages, Filtered & Monitor-Only messages, or Selected filters only.
77. Enter mail.smtp2go.com, port 465, select SSL, enable authentication, and enter the SMTP user name and SMTP password - not the website account password.
78. Enter a verified From address and one or more intended To addresses.
79. Select the minimum subject/body fields required and choose Test.
80. For Selected filters only, edit each intended filter and select Send email.

The provider's current host, port and encryption combinations are documented in [SMTP2GO SMTP Settings](#).

Email privacy: Email can leave the controlled network, remain in mailboxes and appear in notifications. Prefer filtered messages, minimise fields and use an organisation-approved relay and recipient list.

Apprise filtered-message push

Figure 17. Apprise dialog with fictional masked API settings.

Apprise is independent of email. Every non-rejected, non-duplicate message that reaches the lower filtered output can be queued for Apprise. Unfiltered and Monitor Only records are not pushed. By default the body says only A filtered message was received. Filter details and decoded text require explicit opt-in and may appear on a phone lock screen.

Operate an Apprise API

PDW expects an operator-managed Apprise API. Deployment instructions and the container are maintained at [the official Apprise API project](#).

- 81.The beta client contract is Apprise API 1.5.1 with Apprise 1.12.0. Pin those versions or regression-test a newer release before changing it.
- 82.Place the API behind HTTPS and a reverse proxy that enforces Basic authentication. Do not expose the default container port to an untrusted network.
- 83.Use /notify for stateless delivery and supply one or more Destination URLs in PDW.
- 84.Use /notify/{key} for destinations stored on the server and leave PDW Destination URLs blank.
- 85.The endpoint, destinations and API credentials are stored in Windows Credential Manager and masked in PDW.

Pushover phone setup

Create a user account at [Pushover signup](#), install the phone app, and register a PDW application/API token in the dashboard.

Use the account User Key and application token in the Apprise destination format:

```
pover://USER_KEY@APPLICATION_TOKEN
```

Pushover's account keys and application-token process are described in the [official Pushover Message API guide](#).

ntfy phone setup

Install the Android/iOS app or use the web app described in the [official ntfy getting-started guide](#).

For the public HTTPS service, use a long random topic and an Apprise URL such as:

```
ntfys://ntfy.sh/pdw-training-7fc2e91a-example
```

Public ntfy.sh topics are open by default: anyone who knows the topic can subscribe. For sensitive use, reserve/authenticate the topic where available or operate a private ntfy server with access control.

Current secure ntfys syntax and authentication forms are documented by [Apprise ntfy service documentation](#).

Test sequence

86. Enter the authenticated HTTPS Apprise API URL and the correct stateless or stateful destination arrangement.
87. Choose Send test notification. A test does not enable automatic delivery or save unsaved credentials.
88. Confirm the device receives PDW Apprise test.
89. Enable Apprise and save.
90. Replay or inject a synthetic record that matches a training filter; confirm one notification.
91. Replay a synthetic unfiltered record; confirm no filtered-only notification.
92. Review Apprise.log for sanitized result categories if delivery fails.

10. Secure file transfer

The screenshot shows the 'Continuous File Transfer' dialog box with the following configuration:

- Connection:**
 - Protocol: SFTP
 - Server: sftp.training-host.example
 - Port: 22
 - Username: pdw_training
 - Password: [masked]
 - Clear button
 - Remote folder: /public_html/pdw
 - ☒ Passive mode
 - SSH host key SHA256: SHA256:Vv8jQ6vXq1J6mAEoZCsIK8kUfjQHbVX1CjB6hEm9rl0
 - Instruction: SFTP encrypts the connection. Paste the provider's SHA256 SSH host-key fingerprint to prevent impersonation.
- Files to upload:**
 - File list: C:\PDW-Training\Published\index.html, C:\PDW-Training\Published\messages.json, C:\PDW-Training\Published\messages.rss
 - Add files... button
 - Remove button
 - Instruction: Each file is uploaded using its existing filename.
- Schedule:**
 - ☒ Enable automatic uploads
 - Every 60 seconds (10 to 86400)
- Status:**
 - FTP, FTPS and SFTP uploader is ready.
- Buttons:** Upload now, OK, Cancel

Figure 18. SFTP upload profile using fictional host and files.

Protocol choices

Protocol	Usual port	Protection	Use
FTP	21	None	Legacy only. Username, password and files travel in plain text.
FTPS explicit	21	TLS upgrade	Choose only when the host specifies explicit FTPS.
FTPS implicit	990	TLS from connection start	Choose only when the host specifies implicit FTPS.
SFTP	22	SSH	Preferred when the host provides password SFTP and a verified SHA-256 host key.

Get a hosting account

Choose a provider that explicitly supports password-authenticated SFTP or FTPS, supplies a public web folder when you are publishing a site, and can provide the server's verified SSH host-key fingerprint. PDW does not create remote folders.

One example is a conventional web-hosting account such as [DreamHost web hosting](#). This is an example, not an endorsement; compare privacy, location, support and cost for your use.

DreamHost's current SFTP credential guide shows where to obtain hostname, username, password and port 22: [FTP overview and credentials](#). Obtain the SHA-256 host key separately from the provider through a trusted channel.

Configure an upload

93. Open Settings > Connections and automation > File transfer.
94. Choose the exact protocol named by the provider. FTPS and SFTP are different protocols.
95. Enter only the server hostname, then confirm the port, username and provider-specific remote folder.
96. For SFTP, paste the provider's SHA256: host-key fingerprint. Do not copy it from an unexpected first-connection warning.
97. Use Add files to select outputs. PDW rejects duplicate filenames because both would target the same remote name.
98. Set an interval from 10 to 86400 seconds and select Enable automatic uploads.
99. Choose Upload now for an immediate background transfer, then review the status and FileTransfer.log.

Security behaviour

100. The hosting password is a generic Windows Credential Manager entry scoped to the PDW installation; it is not written to PDW.INI or FileTransfer.log.
101. FTPS requires TLS 1.2 or later and validates certificate trust and hostname. There is no bypass switch.
102. SFTP authenticates only after the exact SHA-256 SSH host key matches.
103. Uploads use a temporary local snapshot so a changing logfile is transferred as a consistent file.
104. One background upload runs at a time; the next interval begins after it finishes.
105. The remote directory must exist and file names are retained.

Host-key rotation: If a provider changes its SSH host key, stop transfers and verify the new fingerprint with the provider through a separate trusted channel before updating PDW.

11. Publish to a website or webhook

Figure 19. Publishing profile with conservative privacy settings and fictional webhook values.

Acknowledgement required: Publishing remains off until the operator acknowledges responsibility for checking permissions, privacy obligations, radio laws and publication laws in their jurisdiction. The acknowledgement is not legal advice or permission.

Start with a privacy-minimised profile

106. Filtered messages only.
107. Mask pager address except the last three digits.
108. Leave Include decoded message text off.
109. Use a neutral public source alias that does not expose a site, person or receiver location.
110. Publish first to a private test destination using synthetic records.

Static files

File	Purpose
messages.json	Most recent events as a JSON feed.

File	Purpose
messages.jsonl	One JSON event per line for scripts or imports.
messages.rss	RSS 2.0 feed.
messages.atom	Atom feed.
index.html	Responsive standalone browser table.

Select a local folder, then point File transfer at the generated files to place them on an SFTP/FTPS web host. A local web server can also serve the folder directly when properly secured.

Generic HTTPS webhook

PDW sends one JSON object per event and refuses non-HTTPS URLs. Optional bearer and HMAC secrets are stored in Windows Credential Manager. An Idempotency-Key identifies the event. The receiving service must actually validate the bearer token or X-PDW-Signature; entering a secret in PDW alone does not secure an endpoint.

Pipedream synthetic test example

Create a free account from the [Pipedream getting-started page](#), then create a workflow with an HTTP / Webhook trigger.

111. Copy the generated HTTPS endpoint into PDW's URL field.
112. Keep message text off and use only synthetic records during initial testing.
113. If using a bearer token or HMAC secret, add workflow logic that rejects missing or invalid headers before any downstream step.
114. Choose Test webhook. PDW sends a configuration-only test object without decoded pager text.
115. Inspect the event, then delete synthetic test events from the service if they are no longer required.
116. Enable live delivery only after reviewing the provider's storage, retention, region, access controls and account security.

Pipedream explains generated HTTP endpoints in its [HTTP trigger documentation](#).

Third-party storage: A webhook provider may retain request bodies, headers and event history. Do not send live decoded content until the service and workflow have been approved for that data.

Queue and failures

117. Delivery is asynchronous and does not wait in the capture/decoder thread.
118. Pending events are stored under PublishQueue. Pause retains queued work.
119. Transient failures retry up to five times with exponential backoff.
120. Repeated failures move under PublishQueue\DeadLetter for operator review.
121. The minimum interval provides simple rate limiting.
122. Privacy transforms change only the published copy; the display, logs, email and Apprise source record are not rewritten.

12. Daily operation and troubleshooting

Start-of-shift checklist

123. Confirm the instance name, decoder mode and intended signal source.
124. Confirm receiver/test status and the signal indicator without exposing live content in screenshots.
125. Check date, time and available disk space.
126. Confirm the correct log and filter files are open.
127. Review Apprise, transfer and publishing status only if those channels are approved and enabled.
128. Check PublishQueue\DeadLetter and recent sanitized logs for unresolved failures.
129. Run a synthetic filtered/unfiltered test after material configuration changes.

Troubleshooting

Symptom	Checks and corrective action
No signal movement	Confirm source, cable/device, Windows input permission, selected soundcard/COM port, receiver activity and volume/gain. Test the RTL source where applicable.
Activity but no messages	Confirm mode, frequency, PPM, bandwidth and expected rate. Reduce clipping, check antenna and try known-good permitted traffic.
Poor local-audio decode	Try the supplied audio presets and level changes before Custom. Record a short authorised sample and compare metrics.
Direct RTL-SDR not listed	Close other SDR software, reconnect the device, confirm WinUSB on the exact interface, select the bundled pack and verify a compatible 32-bit DLL.
RTL-TCP test fails	Confirm the host resolves, port is listening, firewall permits it and the server returns a valid RTL0 header. Use a trusted network.
Replay will not start	Stop diagnostic recording, confirm supported WAV/SigMF encoding and path permissions, then retry.
Expected filter does not match	Check type, address length/wildcards, function number, all & fragments, exact-match setting and Reject/Monitor Only state.
Email test fails	Verify provider host, implicit SSL port, SMTP credentials, verified From address and network access. Use the relay's SMTP user password.
No Apprise notification	Confirm the record reaches the lower filtered pane, Apprise is enabled, HTTPS/Basic auth works, and stateless destinations are not mixed with /notify/{key}.
SFTP upload fails	Confirm host, port 22, password, existing remote folder and exact provider-verified SHA-256 host key.
FTPS certificate error	Correct hostname and system trust/time. Do not bypass certificate validation.
Publishing test disabled/fails	Enable a static or webhook output, acknowledge jurisdiction responsibility, use HTTPS and check pause/rate controls.
Queue grows	Pause new exposure if needed, correct the destination, review pending/dead-letter data and re-enable only after a synthetic test.
Second PDW instance will not start	Use a separate complete folder for each concurrent instance and give it a unique WindowTitle.

Diagnostic logs

Log / folder	Use	Sensitive content
FileTransfer.log	Transfer result and timing review.	Designed to exclude the password; uploaded filenames

Log / folder	Use	Sensitive content
		and operational details still need protection.
Apprise.log	Timestamp, generated event identifier and sanitized result category.	Excludes decoded content, endpoint, credentials, destination URL and response body.
Published	Inspect generated static outputs before upload.	May contain published addresses, labels and message text according to the profile.
PublishQueue	Pending webhook events.	Contains event payloads; protect it as decoded data.
DeadLetter	Repeated delivery failures.	Contains failed event payloads; review and delete under policy.

Safe acceptance test

130. Copy the PDW folder to an isolated training location.
131. Use a synthetic or explicitly licensed recording with one known matching and one known non-matching event.
132. Confirm display fields, filter labels, hit counters and local logs.
133. Use disposable third-party test destinations and no real tokens in screenshots or issue reports.
134. Confirm filtered-only outputs send the matching record once and do not send the non-matching record.
135. Deliberately test a wrong SFTP host key and an invalid TLS endpoint; both should fail closed.
136. Remove disposable credentials and test data after acceptance.

Backup and restore

137. Close PDW before backing up settings.
138. Back up PDW.INI, filters.ini, required WAV/database files and receiver packages you intentionally imported.
139. Windows Credential Manager secrets are not present in the folder backup; re-enter them on the restored machine.
140. Do not copy private logs, recordings, Published or PublishQueue folders unless the backup is authorised and encrypted.
141. Restore into a separate folder first, verify settings, then retire the old copy only after acceptance.

13. Reference

Menu reference

Menu	Commands
File	Log file; Save copied data; Print copied data; Exit
Edit	Copy Selection; Copy monitor window; Copy filtered window; Clear monitor
Monitor	Input setup; Volume mixer; Decoding mode; Statistics; Pause or resume
Filters	Manage filters; Filter options; Reload filters; Reset all hit counters; Write filter file; Run filter command file
Settings	Settings hub; Appearance; Decoder and input; Connections and automation; Character set
Help	User guide; Diagnostics; About

Keyboard shortcuts

Shortcut	Command	Shortcut	Command
Alt+L	Log file	Ctrl+C	Copy selection
Ctrl+U	Copy monitor	Ctrl+L	Copy filtered
Ctrl+D	Clear display	Ctrl+F	Manage filters
Ctrl+R	Reload filters	Ctrl+,	Settings hub
Ctrl+G	General behaviour	Ctrl+O	Decoder options
Ctrl+S	Input setup	Ctrl+M	Email
Ctrl+T	System tray	Alt+V	Volume mixer
Alt+S	Statistics	Alt+F	Message font
Ctrl+P	Print copied data	Ctrl+X	Exit
F1	User guide	F11	Switch pane sizes
F12	Diagnostics	Alt+Shift+R	Legacy recording
Alt+Shift+P	Legacy playback	Alt+Shift+A	Legacy auto-record

Radio defaults and limits

Setting	Default / range
RTL-TCP host	127.0.0.1
RTL-TCP port	1234
Frequency	148000000 Hz initial value - replace with an authorised target
IQ sample rate	1024000 samples/s
Audio rate	48000 samples/s
NFM bandwidth	12000 Hz default; 5000 to 25000 Hz

Setting	Default / range
PPM correction	0 initial
Automatic gain	Enabled initial
Recording safety limit	25 million normalized samples in memory
Calibration search	1,000 threshold/centering/resync combinations

Third-party service directory

Service	Why needed	Key setup output
SMTP2GO	Email relay account and SMTP user	SMTP host mail.smtp2go.com; PDW example uses SSL port 465.
Apprise API	Operator-managed push gateway	Host separately behind HTTPS and Basic authentication.
Pushover	Phone/desktop push destination	Account User Key plus an application API token.
ntfy	Phone/web push destination	Public or self-hosted topic; use secure ntfys URL and access controls.
Pipedream	Optional webhook workflow	Synthetic testing or reviewed automation; validate bearer/HMAC before downstream actions.
SFTP web host	Static website/file destination	Must provide credentials, remote folder and verified host key.
Zadig	Optional WinUSB driver utility	No account. Use only on the exact RTL-SDR interface.

Official links

[PDW project site](#)

[PDW source repository](#)

[Apprise API](#)

[Apprise supported services](#)

[Pushover signup](#)

[Pushover Message API](#)

[ntfy getting started](#)

[SMTP2GO signup](#)

[SMTP2GO SMTP settings](#)

[Pipedream HTTP triggers](#)

[DreamHost SFTP credentials example](#)

[Zadig](#)

[RTL-SDR Blog quick start](#)

Glossary

Term	Meaning
------	---------

Term	Meaning
Capcode / Riccode	Address used by a paging receiver and filter.
Discriminator tap	Receiver output taken before normal audio filtering/de-emphasis for a cleaner data waveform.
FTPS	FTP protected by TLS. It is not SFTP.
SFTP	SSH File Transfer Protocol.
Host key	SSH server identity fingerprint verified before password authentication.
HMAC	Message authentication code computed over the exact webhook body with a shared secret.
Idempotency key	Stable event identifier a receiver can use to avoid processing the same event twice.
IQ samples	In-phase and quadrature radio samples received from an SDR.
NFM	Narrowband frequency modulation.
SigMF	Signal Metadata Format recording represented here as rf32_le data plus metadata.
WASAPI	Current Windows audio capture API used as a fallback beside retained WinMM.
Dead letter	Repeatedly failed queued event retained for operator review.

Credits and licence

PDW credits retained from the application:

- 142. Jason Petty - original PDW project.
- 143. Peter Hunt - PDW development from 2004 to 2010.
- 144. Open-source release contributors, including later build and compatibility work.
- 145. Kieran O'Rourke - PDW v4.1.0 beta modernization contribution credited by the current About dialog.

The supplied application identifies the GNU General Public License, version 3 or later. Review the included License and THIRD_PARTY_NOTICES.md files for the complete terms, bundled component versions, sources and licences.

Manual maintenance: This edition reflects PDW v4.1.0 Beta and third-party documentation checked in August 2026. Re-check service URLs, account requirements, limits and security guidance before a later release or production rollout.